Outcome Imputation in `predict.crc_fit()`

Internal mathematical note

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1 Purpose and scope

This note describes the mathematical calculations used by `predict.crc_fit()` to estimate the total value of the modeled outcome. The method combines conditional imputation for observed units with prediction for units belonging to the all-zero capture cells. It applies to exact, interval-censored, right-censored, and missing positive count outcomes and to both outcome distributions supported by the package.

The calculation is performed after the EM algorithm has converged. All model parameters and posterior state weights below are therefore evaluated at their final fitted values. Hats are omitted where doing so does not cause ambiguity.

2 State and outcome notation

Let $i \in S$ index an observed unit and let $s \in \mathcal{S}_i$ index a candidate state of that unit. A state records the candidate true values of misclassified categorical variables and the latent class. Write

$$\tau_{is} = \Pr(s \mid \text{observed information for unit } i)$$

for its final posterior weight. These weights satisfy

$$\tau_{is} \geq 0, \qquad \sum_{s \in \mathcal{S}_i} \tau_{is} = 1.$$

For state s, the outcome regression provides the linear predictor and mean

$$\eta_{is} = z_{is}^\top \beta, \qquad \mu_{is} = \exp(\eta_{is}).$$

The parameter μ_{is} is the mean of an underlying untruncated count W. The modeled positive outcome is

$$V = W \mid W \geq 1.$$

The underlying distribution is either

$$W \sim \text{Poisson}(\mu)$$

or

$$W \sim \text{NB}(\mu, r), \qquad r = \frac{1}{\sigma},$$

under the mean–size parameterization

$$\text{E}(W) = \mu, \qquad \text{Var}(W) = \mu + \sigma\mu^2.$$

Define the survival function

$$S(q; \mu) = \Pr(W > q \mid \mu).$$

The implementation evaluates $\log S(q; \mu)$ directly through the upper-tail interfaces of the Poisson and negative-binomial distribution functions.

3 Shifted-distribution identities

Conditional means over censored ranges can be evaluated without explicitly summing $w \Pr(W = w)$. For a Poisson random variable,

$$w \Pr(W = w; \mu) = \mu \Pr(W^\star = w - 1; \mu),$$

where $W^\star \sim \text{Poisson}(\mu)$. Consequently,

$$\sum_{w=l}^{u} w \Pr(W = w) = \mu \Pr(l - 1 \le W^\star \le u - 1).$$

For a negative-binomial random variable with size r and mean μ,

$$w \Pr(W = w; \mu, r) = \mu \Pr(W^\star = w - 1; \mu^\star, r^\star),$$

where

$$r^\star = r + 1, \qquad \mu^\star = \mu \frac{r + 1}{r}.$$

This choice preserves the probability parameter of the negative-binomial distribution. The implementation calls this the *shifted* distribution. Let $S^\star(q; \mu)$ denote its survival function. For the Poisson model, $S^\star$ has the same parameters as S; for the negative-binomial model, it uses $(\mu^\star, r^\star)$.

The preceding identities imply

$$\begin{aligned} \mathrm{E}\{W\mathbf{1}(W \ge l)\} &= \mu S^\star(l - 2; \mu), \\ \mathrm{E}\{W\mathbf{1}(l \le W \le u)\} &= \mu\{S^\star(l - 2; \mu) - S^\star(u - 1; \mu)\}. \end{aligned}$$

4 Conditional imputation for observed units

For every observed unit and candidate state, the function constructs

$$v_{is}^{\text{imp}} = \mathrm{E}(V_i \mid \mathcal{O}_i, s),$$

where $\mathcal{O}_i$ denotes the exact, censored, or missing outcome information available for unit i.

4.1 Missing outcome

When the outcome is missing, the only outcome information is the positive support of V. Since the zero value contributes nothing to $\mathrm{E}(W)$,

$$\mathrm{E}(V \mid \mu) = \mathrm{E}(W \mid W \ge 1) = \frac{\mu}{\Pr(W \ge 1)} = \frac{\mu}{S(0; \mu)}.$$

The implementation evaluates this as

$$\exp\{\log \mu - \log S(0; \mu)\},$$

which avoids overflow caused by separately calculating the reciprocal of a very small survival probability.

4.2 Exact outcome

For an exact observed value v_i,

$$v_{is}^{\text{imp}} = \mathrm{E}(V_i \mid V_i = v_i, s) = v_i.$$

This value is independent of the candidate state. Consequently, posterior averaging preserves the observation because

$$\sum_{s \in \mathcal{S}_i} \tau_{is} v_i = v_i.$$

4.3 Right-censored outcome

Suppose the available information is $V_i \geq l_i$. Because $l_i \geq 1$, conditioning additionally on zero truncation is redundant. The conditional mean is

$$\begin{aligned} \mathrm{E}(W \mid W \geq l) &= \frac{\mathrm{E}\{W\mathbf{1}(W \geq l)\}}{\Pr(W \geq l)} \\ &= \mu \frac{S^\star(l-2;\mu)}{S(l-1;\mu)}. \end{aligned}$$

The log-scale expression used in the function is

$$\exp\{\log \mu + \log S^\star(l-2;\mu) - \log S(l-1;\mu)\}.$$

When $l = 1$, $S^\star(-1;\mu) = 1$, and the result reduces to the ordinary zero-truncated mean.

4.4 Interval-censored outcome

Suppose that $l_i \leq V_i \leq u_i$, with both endpoints included. The denominator of the conditional expectation is

$$\Pr(l \leq W \leq u) = S(l-1;\mu) - S(u;\mu).$$

Using the shifted-distribution identity, its numerator is

$$\mathrm{E}\{W\mathbf{1}(l \leq W \leq u)\} = \mu\{S^\star(l-2;\mu) - S^\star(u-1;\mu)\}.$$

It follows that

$$\mathrm{E}(W \mid l \leq W \leq u) = \mu \frac{S^\star(l-2;\mu) - S^\star(u-1;\mu)}{S(l-1;\mu) - S(u;\mu)}.$$

Both probability differences are evaluated on the log scale. If $a = \log A$ and $b = \log B$ with $A \geq B$, then

$$\log(A - B) = a + \log\{-\operatorname{expm1}(b-a)\}.$$

This is the operation implemented by `log_sub_crc()`. It avoids catastrophic cancellation when two right-tail probabilities are close.

5 Observed-unit contribution

Each expanded state represents a fractional expected unit with population contribution τ_{is}. Its estimated outcome contribution is

$$\tau_{is} v_{is}^{\text{imp}}.$$

Summing over states and observed units gives

$$\widehat{T}_{\text{obs}} = \sum_{i \in S} \sum_{s \in \mathcal{S}_i} \tau_{is} v_{is}^{\text{imp}}.$$

The posterior weights already incorporate the capture model, misclassification probabilities, and outcome likelihood. Thus, imputation for a censored or missing outcome is averaged over uncertainty about its true categorical values and latent class.

6 Contribution from completely unobserved units

Let c index a complete capture cell with the all-zero capture history. The fitted capture model supplies its expected population count

$$m_c = \exp(x_c^\mathsf{T} \theta).$$

All categorical variables required by the outcome model are present in the capture-cell grid. Therefore, the outcome design row z_c and untruncated outcome mean

$$\mu_c = \exp(z_c^\mathsf{T} \beta)$$

can be constructed for every all-zero cell. Since no outcome is observed for these units, their expected positive outcome is

$$\mathrm{E}(V \mid c) = \frac{\mu_c}{S(0; \mu_c)}.$$

The estimated outcome total contributed by cell c is consequently

$$m_c \frac{\mu_c}{S(0; \mu_c)}.$$

The implementation rejects non-finite or non-positive values of μ_c and evaluates the zero-truncated mean on the log scale.

7 Overall and subpopulation estimators

Combining observed and completely unobserved contributions gives

$$\widehat{T} = \widehat{T}_{\text{obs}} + \sum_{c:\, \boldsymbol{y}_c = \mathbf{0}} m_c \frac{\mu_c}{S(0; \mu_c)}.$$

The corresponding population-size estimator is

$$\widehat{N} = \sum_{i \in S} \sum_{s \in \mathcal{S}_i} \tau_{is} + \sum_{c:\, \boldsymbol{y}_c = \mathbf{0}} m_c = |S| + \sum_{c:\, \boldsymbol{y}_c = \mathbf{0}} m_c.$$

For a subpopulation A defined by categorical variables, the latent class, or both, the function restricts the preceding sums to states and all-zero cells belonging to A:

$$\begin{aligned}
\widehat{N}_A &= \sum_{i \in S} \sum_{s \in \mathcal{S}_i} \tau_{is} \mathbf{1}(s \in A) + \sum_{c:\, \boldsymbol{y}_c = \mathbf{0}} m_c \mathbf{1}(c \in A), \\
\widehat{T}_A &= \sum_{i \in S} \sum_{s \in \mathcal{S}_i} \tau_{is} v_{is}^{\text{imp}} \mathbf{1}(s \in A) + \sum_{c:\, \boldsymbol{y}_c = \mathbf{0}} m_c \frac{\mu_c}{S(0; \mu_c)} \mathbf{1}(c \in A).
\end{aligned}$$

These subgroup estimates add to the corresponding overall estimates when the grouping variables define a partition of the modeled population.

8 Summary of the implemented cases

Outcome information	Imputed state value
Missing	$\mu/S(0;\mu)$
Exact v	v
Right-censored at l	$\mu S^{\star}(l-2;\mu)/S(l-1;\mu)$
Interval $[l,u]$	$\mu\{S^{\star}(l-2;\mu)-S^{\star}(u-1;\mu)\}/\{S(l-1;\mu)-S(u;\mu)\}$